ANALYSIS REPORT

Track 1 Extended Overnight Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the extended `Track 1` campaign prepared in:

- `doc/reports/campaign_plans/2026-04-13-13-27-37_track1_extended_overnight_campaign_plan_report.md`

  The package executed `48` repository-owned harmonic-wise validation runs through one coordinated launcher:
- completed runs: `48`
- failed runs: `0`
- execution window: `2026-04-13T15:10:50+02:00` to `2026-04-13T16:00:30+02:00`
- campaign artifact root:

  `output/training_campaigns/track1_extended_overnight_campaign_2026_04_13_13_31_57/` The extended batch was organized into six logical experiment blocks:
- `Campaign A` : wide low-order `HGBM` search
- `Campaign B` : heavy low-order `HGBM` escalation
- `Campaign C` : late-harmonic repair `HGBM`
- `Campaign D` : bridge joint `HGBM`
- `Campaign E` : engineered-feature `HGBM` controls
- `Campaign F` : `RandomForest` controls

Objective And Outcome

The campaign had four practical questions

1. can a much wider low-order search finally beat the current `Track 1` shared-evaluator reference;
2. can a heavier low-order budget outperform the lighter winners when the budget is increased aggressively;
3. can bridge runs that couple `h0/h1` with the late-harmonic cluster close the remaining gap better than isolated ladders;
4. do engineered terms or `RandomForest` recover enough value to re-enter the promotion discussion under a much broader budget.

   Outcome:

- the batch completed successfully with `48/48` runs and no observed launcher failures;
- the campaign winner is `track1_hgbm_h01_wide_depth_2` ;
- the best test mean percentage error improved from the previous repository best `8.774%` to `8.707%` ;
- the improvement is real but still small, so `Target A` remains `not_yet_met` against the paper threshold `4.7%` ;
- the strongest alternatives are still `HGBM` , especially the bridge joint and late-repair blocks;
- the heavy low-order escalation did not beat the lighter wide winner, which means extra budget alone is not enough;
- engineered-term and `RandomForest` controls remained clearly non-promotable in this branch.

Ranking Policy

The campaign winner uses the explicit ranking rule approved in the campaign plan:

- eligible winner set: all completed runs that emit the shared offline evaluator;
- primary metric: `test_mean_percentage_error_pct`;
- first tie breaker: `test_curve_mae_deg`;
- second tie breaker: `test_curve_rmse_deg`;
- third tie breaker: lexical `run_name`;
- direction: minimize.

This policy is serialized under:

- `output/training_campaigns/track1_extended_overnight_campaign_2026_04_13_13_31_57/campaign_leaderboard.yaml`

Ranked Completed Runs

Rank	Config	Block	Test MPE [%]	Curve MAE [deg]	Curve RMSE [deg]
1	track1_hgbm_h01_wide_depth_2	A	8.707	0.002611	0.002810
2	track1_hgbm_h01_h162240_joint_balanced	D	8.720	0.002605	0.002802
3	track1_hgbm_h01_wide_depth_1	A	8.759	0.002614	0.002809
4	track1_hgbm_h01_h156162240_joint	D	8.768	0.002648	0.002844
5	track1_hgbm_h0181_h162240_joint	D	8.770	0.002647	0.002843
6	track1_hgbm_h014078_h162240_joint	D	8.772	0.002648	0.002844
7	track1_hgbm_h81156162240_cluster	C	8.778	0.002594	0.002790
8	track1_hgbm_h014078_wide_anchor_1	A	8.797	0.002648	0.002843
9	track1_hgbm_h0139_wide_anchor_1	A	8.797	0.002648	0.002843
10	track1_hgbm_h01_long_budget_regularized	A	8.807	0.002620	0.002817
11	track1_hgbm_h162_h240_wide_repair_1	C	8.914	0.002619	0.002815
12	track1_hgbm_h156162240_bridge_1	C	8.971	0.002632	0.002830
13	track1_hgbm_h01_ultraheavy_guarded	B	8.977	0.002685	0.002881
14	track1_hgbm_h081162240_bridge_1	C	8.986	0.002633	0.002831
15	track1_hgbm_h162_h240_wide_repair_2	C	9.000	0.002633	0.002835
16	track1_hgbm_h0181_wide_anchor_1	A	9.009	0.002677	0.002874
17	track1_hgbm_h01_high_bin_count	A	9.011	0.002678	0.002874
18	track1_hgbm_h013981_wide_anchor	A	9.030	0.002662	0.002860
19	track1_hgbm_h01_wide_depth_3	A	9.032	0.002686	0.002882
20	track1_hgbm_h39162240_bridge_1	C	9.036	0.002664	0.002860
21	track1_hgbm_h01_small_leaf_extreme	A	9.125	0.002700	0.002894
22	track1_hgbm_h013_h162240_joint_heavy	D	9.127	0.002734	0.002927
23	track1_hgbm_h013_h081162240_joint	D	9.129	0.002734	0.002927
24	track1_hgbm_h013_wide_support_2	A	9.131	0.002688	0.002884
25	track1_hgbm_h0181_heavy_reference	B	9.131	0.002735	0.002929
26	track1_hgbm_h014078_heavy_reference	B	9.135	0.002736	0.002930
27	track1_hgbm_h01_heavy_reference	B	9.136	0.002736	0.002930
28	track1_hgbm_h013_h162240_engineered_joint	E	9.155	0.002627	0.002830
29	track1_hgbm_h013_wide_support_1	A	9.177	0.002727	0.002923
30	track1_hgbm_h013_heavy_reference	B	9.199	0.002728	0.002924
31	track1_hgbm_h0139_heavy_reference	B	9.201	0.002728	0.002924
32	track1_hgbm_h013981_h162240_joint	D	9.213	0.002730	0.002926
33	track1_hgbm_h0139_h162240_joint_heavy	D	9.217	0.002732	0.002927
34	track1_hgbm_h162_isolation_heavy	C	9.301	0.002708	0.002904
35	track1_hgbm_h240_isolation_heavy	C	9.302	0.002708	0.002905
36	track1_hgbm_h01_engineered_heavy	E	9.360	0.002677	0.002879
37	track1_hgbm_h013_engineered_heavy	E	9.361	0.002677	0.002878
38	track1_hgbm_h0139_h162240_engineered_joint	E	9.382	0.002650	0.002853
39	track1_hgbm_h162240_engineered_heavy	E	9.644	0.002696	0.002900
40	track1_rf_h01_h81_engineered_control	F	10.982	0.003239	0.003431
41	track1_rf_h81_heavy_control	F	11.200	0.003315	0.003500
42	track1_rf_h039_h162240_bridge_control	F	11.212	0.003317	0.003502
43	track1_rf_h162240_heavy_control	F	11.212	0.003317	0.003502
44	track1_rf_h013_h162240_joint_control	F	11.233	0.003319	0.003504
45	track1_rf_h013981_bridge_control	F	11.233	0.003319	0.003504
46	track1_rf_h01_h81_bridge_control	F	11.236	0.003319	0.003504

Rank	Config	Block	Test MPE [%]	Curve MAE [deg]	Curve RMSE [deg]
47	track1_rf_full_rcim_heavy_control	F	11.244	0.003320	0.003505
48	track1_rf_h01_heavy_control	F	11.244	0.003320	0.003505

Config	Validation MPE [%]	Oracle Test MPE [%]	Delta Vs 8.774% Baseline
track1_hgbm_h01_wide_depth_2	9.830	2.749	-0.067
track1_hgbm_h01_h162240_joint_balanced	10.067	2.749	-0.054
track1_hgbm_h01_wide_depth_1	9.711	2.749	-0.015
track1_hgbm_h01_h156162240_joint	9.621	2.749	-0.006
track1_hgbm_h0181_h162240_joint	9.619	2.749	-0.004

Best Run Per Block

Block	Best Config	Test MPE [%]	Interpretation
A	track1_hgbm_h01_wide_depth_2	8.707	The best result came from the wide low-order search, confirming that controlled h0/h1 tuning is still the strongest direction.
B	track1_hgbm_h01_ultraheavy_guarded	8.977	Heavier low-order budgets did not beat the simpler wide winner, so extra depth and budget alone are not the fix.
C	track1_hgbm_h81156162240_cluster	8.778	Late-harmonic repair remains useful, but it still trails the best low-order and joint variants on the shared evaluator.
D	track1_hgbm_h01_h162240_joint_balanced	8.720	Joint low-order plus late-harmonic coupling nearly matched the winner and remains the strongest secondary promotion path.
E	track1_hgbm_h013_h162240_engineered_joint	9.155	Engineered terms stayed clearly behind the best no-engineering HGBM candidates and do not justify default promotion.
F	track1_rf_h01_h81_engineered_control	10.982	RandomForest remained non-competitive even under the broader and heavier campaign budget.

Campaign Winner

The explicit campaign winner is

- `track1_hgbm_h01_wide_depth_2`

  Its result was:

- test mean percentage error: `8.707%`
- test curve MAE: `0.002611 deg`
- test curve RMSE: `0.002810 deg`
- validation mean percentage error: `9.830%`
- oracle test mean percentage error: `2.749%`
- previous shared-evaluator best: `8.774%`
- absolute improvement versus previous best: `0.067 percentage points`
- Target A threshold: `4.7%`
- Target A status: `not_yet_met`

  This run won because it is the lowest-error completed candidate under the shared offline evaluator, and it beat both the previous repository best and the strongest bridge-joint alternative.

Interpretation By Experiment Block

1. Wide Low-Order Search Won Again

The winner still came from the low-order `HGBM` search, not from the heavy escalation block.

Important evidence

- `track1_hgbm_h01_wide_depth_2` : `8.707%`
- `track1_hgbm_h01_h162240_joint_balanced` : `8.720%`
- `track1_hgbm_h01_wide_depth_1` : `8.759%`
- `track1_hgbm_h01_ultraheavy_guarded` : `8.977%`

Interpretation:

- `h0 / h1` remain the main leverage point for TE-level improvement;
- broader exploration still works better than simply making the same low-order family heavier;
- the campaign therefore does not justify promoting the heavy block as the new default search path.

2. Joint Low-Order Plus Late-Harmonic Coupling Is The Best Secondary Path

The strongest non-winning direction came from the bridge-joint block:

- `track1_hgbm_h01_h162240_joint_balanced` : `8.720%`

Interpretation:

- coupling low-order treatment with `162/240` still looks structurally useful;
- the joint path nearly tied the winner and should remain the main companion direction in the next iteration;
- this is stronger evidence than the previous smaller overnight batch because the bridge family survived a much wider search.

3. Late-Harmonic Repair Helps, But It Still Trails The Best Low-Order Path

The best late-repair run was

- `track1_hgbm_h81156162240_cluster` : `8.778%`

Interpretation:

- the late-harmonic cluster remains relevant;
- however, isolated late repair still does not beat the best low-order or bridge-joint candidate;
- this suggests the remaining gain probably needs better coupling, not just stronger isolated repair.

4. Engineered Terms And RandomForest Still Do Not Earn Promotion

The best engineered and `RandomForest` controls were:

- engineered `HGBM` : `track1_hgbm_h013_h162240_engineered_joint` at `9.155%`
- best `RF` : `track1_rf_h01_h81_engineered_control` at `10.982%`

  Interpretation:
- engineered operating-condition terms remain weaker than the strongest

  no-engineering `HGBM` path;
- `RandomForest` remained far from competitive even after the campaign budget

  was widened and deepened;
- the next `Track 1` push should not spend primary budget on either of these directions.

Recommended Next Step

The batch improved the shared evaluator, but the gain is still incremental.

The most defensible next step is

1. promote `track1_hgbm_h01_wide_depth_2` as the new shared-evaluator reference;
2. keep `track1_hgbm_h01_h162240_joint_balanced` as the strongest bridge-joint companion;
3. keep `track1_hgbm_h81156162240_cluster` as the strongest late-repair companion;
4. design the next `Track 1` iteration around a combined low-order plus

   late-harmonic coupling path instead of another broad engineered or `RF` retry;
5. if this stronger coupling path still stalls near `8.7%`, move the next

   research step to a new target-parameterization implementation rather than only adding more budget to the current coefficient path.

Artifact References

- `output/training_campaigns/track1_extended_overnight_campaign_2026_04_13_13_31_57/campaign_leaderboard.yaml`
- `output/training_campaigns/track1_extended_overnight_campaign_2026_04_13_13_31_57/campaign_best_run.yaml`
- `output/training_campaigns/track1_extended_overnight_campaign_2026_04_13_13_31_57/campaign_best_run.md`
- `output/validation_checks/paper_reimplementation_rcim_harmonic_wise/2026-04-13-15-11-49__track1_hgbm_h01_wide_depth_2_campaign_run/validation_summary.yaml`
- `doc/reports/analysis/Training Results Master Summary.md`
- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`